

# Noubtigo – Product Requirements Document (PRD)

## Product Overview

Noubtigo is a multi-tenant SaaS platform for queue management, appointment scheduling, customer tracking, and waiting room optimization.

The platform is designed for:

- Clinics
- Medical centers
- Hospitals
- Banks
- Government offices
- Automotive services
- Public service organizations
- Any business managing physical customer visits

The system combines traditional queue management with appointment scheduling and real-time customer tracking.

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## Product Goals

The platform aims to:

1. Reduce waiting time.
  2. Improve customer experience.
  3. Allow remote queue tracking.
  4. Reduce crowding in waiting rooms.
  5. Provide analytics and operational insights.
  6. Support businesses of different sizes through subscription plans.
  7. Operate as a scalable SaaS platform.
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## Core Architecture

### Multi-Tenant SaaS

The application uses a multi-tenant architecture.

Each company has isolated:

- Users
- Customers
- Tickets
- Appointments
- Services
- Rooms
- Settings
- Activity Logs
- Reports

All business data must be protected by company-level isolation.

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## User Roles

### Super Admin

Platform owner.

Can:

- Manage companies
- Manage plans
- Manage subscriptions
- Access global reports

### Company Admin

Can:

- Manage company settings
- Manage users
- Manage services
- Manage rooms
- View reports

### Secretary

Can:

- Create customers
- Create tickets
- Create appointments

- Manage queue operations

## **Staff / Operator**

Can:

- Call next customer
  - Serve customers
  - Complete tickets
  - Use queue actions according to permissions
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# **Queue Management**

## **Ticket Creation**

Tickets may be created:

- For existing customers
- For new customers

Each ticket belongs to:

- Company
- Service
- Room

Ticket statuses:

- waiting
  - called
  - serving
  - on\_hold
  - done
  - cancelled
  - no\_show
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## **On Hold / Resume Service**

Staff may temporarily pause a ticket while serving a customer.

Examples:

- Customer needs to complete payment

- Customer needs medical examination
- Missing documents
- Customer temporarily leaves

Actions:

- Put On Hold
- Resume Service

All actions must be logged.

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## **Ticket Cancellation**

Staff may cancel tickets.

Reasons:

- Customer requested cancellation
- Customer left
- Duplicate ticket
- Wrong service
- Other

Cancelled tickets remain stored for reporting and audit purposes.

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## **Reopen Cancelled Ticket**

Authorized staff may reopen cancelled tickets.

Rules:

- Reopened ticket returns to waiting queue.
  - Original ticket is reused.
  - Complete history is preserved.
  - Activity logs must record reopening actions.
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## **Appointment Management**

Features:

- Schedule appointments

- Service-specific time slots
- Capacity management
- Appointment check-in
- Appointment reminders

Appointment statuses:

- scheduled
- checked\_in
- waiting
- serving
- completed
- cancelled
- no\_show

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## Customer Management

Customer fields:

- First Name
- Last Name
- Phone Number
- Email
- Unique Identifier
- Notes

Customer search must support:

- First name
- Last name
- Phone number
- Email
- Identifier

The system must support thousands of customers efficiently using AJAX search.

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## Customer Tracking

Customers can track their position using:

### Method 1

Track by Ticket Number

## Method 2

Track via WhatsApp

## Method 3 (Future)

Customer Account Portal

Features:

- Visit history
  - Favorite companies
  - Notifications
  - PWA support
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## Public Display Screen

Real-time display showing:

- Current ticket
- Next tickets
- Sound notifications

Designed for:

- TVs
  - Waiting rooms
  - Kiosks
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## Real-Time System

Use:

- Node.js
- Socket.io

The architecture should support future scaling and Redis integration.

Redis is not required initially.

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# Activity Logs

All important actions must be logged.

Examples:

- Create
- Update
- Delete
- Login
- Permission changes
- Queue actions
- Appointment actions

Every major entity should have a timeline/history view.

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# Analytics

Provide:

- Average waiting time
- Peak hours
- Service demand
- Staff performance
- Cancellation rate
- Hold duration metrics

Charts:

- Daily
  - Weekly
  - Monthly
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# Subscription System

Plans:

- Free
- Basic
- Pro
- Enterprise

Support:

- Monthly billing
- Annual billing

Current phase:

- Manual payment approval

Future:

- Credit Card
  - Debit Card
  - PayPal
  - Stripe
  - Local payment gateways
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## Multi-Language

Languages:

- English
- French
- Arabic

Requirements:

- RTL support for Arabic
  - Translation files
  - No hardcoded text
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## UI/UX Requirements

Design style:

- Modern SaaS
- Mobile-first
- Responsive
- Fast

Branding inspired by Noubtigo logo colors.

Use:

- Modern cards
  - Clean dashboards
  - Searchable tables
  - Filters
  - AJAX-powered components
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## PWA Support

The application should support:

- Installation on mobile devices
  - Offline-ready architecture for future enhancements
  - Push notifications in future releases
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## Security

Requirements:

- RBAC
  - Tenant isolation
  - CSRF protection
  - XSS protection
  - Rate limiting
  - Activity audit logs
  - Permission-based access control
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## Scalability

The platform must support:

- Thousands of companies
- Multiple branches
- Large customer databases
- High queue volume

Architecture should be prepared for future growth without major redesign.

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# Product Vision

Noubtigo aims to become a complete customer flow and queue management platform that combines queueing, appointments, customer tracking, analytics, and SaaS monetization in a single scalable solution.